5.6 The Fermat Point

WORKSHEET

The question

Given that $\alpha = e^{i\pi/3}$, prove that $1 + \alpha^2 = \alpha$.

A triangle in the Argand plane has vertices A, B, and C represented by the complex numbers p, $q\alpha^2$ and $-r\alpha$ respectively, where p, q and r are positive real numbers. Sketch the triangle ABC.

Three equilateral triangles ABL, BCM and CAN (each lettered clockwise) are erected on sides AB, BC and CA respectively. Show that the complex number representing N is $(1-\alpha)p - \alpha^2 r$ and find similar expressions for the complex numbers representing L and M.

Show that lines LC, MA and NB all meet at the origin, and that these three line segments have the common length $p + q + r$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Where are A, B and C? What are their arguments? What angles do OA, OB and OC make with each other?

STAGE 2 Equilateral triangles as rotations

Ask yourself Triangle CAN is equilateral, lettered clockwise. If you stand at C, how do you get from A to N? Express $N - C$ in terms of $A - C$.

Pause and try Find L and M in the same way, and simplify using the identities from Stage 1.

STAGE 3 Three lines through the origin

Ask yourself Compare $L = \alpha(p + q)$ with $C = -r\alpha$. What do you notice? Do the same for M and A, and for N and B.

Try this next

1. **Check the lettering.** Show that if the equilateral triangles were lettered anticlockwise (built inwards), the three lines would generally not pass through O.

2. **Napoleon's theorem.** Show that the centre of the equilateral triangle ABL is $\frac{1}{3}(A + B + L)$. Using the expressions above, show that the three centres form an equilateral triangle.

3. **The minimum.** With $p = 2$, $q = 1.4$, $r = 1.1$, compute $PA + PB + PC$ for P at each vertex and at the centroid $\frac{1}{3}(A + B + C)$, and check that all exceed 4.5.

4. **An equilateral test.** Show that z_1, z_2, z_3 form an equilateral triangle (in either orientation) if and only if $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$.

Further reading

- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 7 discusses the Fermat point, Steiner trees and soap films.
- **Liang-shin Hahn, *Complex Numbers and Geometry*** (Mathematical Association of America). Many geometric theorems proved with complex numbers, at exactly this

level.

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 1 includes Napoleon's theorem and other geometric results.

The full guided solution, and what the question leads to at university, is in Section 5.6:
https://beyondstep.keithmcnulty.org/ch05/fermat-point.html